SIMATS ENGINEERING

ASSIGNMENT - 2

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO2: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

Assessment Tool Weightage: 35%

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Dept. of CSE

QUESTIONS: 5 **Total Marks: 50**

Annexure A

ASSIGNMENT - 2

Code of Conduct:

I Barath. S (192521359) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Criteria	Weightage (Marks)	Level 4 – Exemplary (Full Marks)	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Total
Conceptual Understanding	35 Marks					
Accuracy of Answers	25 Marks					
Application of Knowledge	15 Marks					
Completeness of Response	15 Marks					
Clarity & Presentation	10 Marks					
Total						

RUBRICS FOR CALCULATION

Criteria	Weightage (Marks)	Level 4 – Exemplary (Full Marks)	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	35 Marks	Demonstrates clear, complete, and in-depth understanding of OS concepts (types, structure, system calls)	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answers	25 Marks	All answers are correct, precise, and relevant	Mostly correct with minor errors	Partially correct with noticeable errors	Mostly incorrect or irrelevant
Application of Knowledge	15 Marks	Effectively applies concepts to scenarios/questions	Applies with minor mistakes	Limited or partial application	No meaningful application
Completeness of Response	15 Marks	All parts of each question are fully answered	Most parts covered with small omissions	Some parts missing	Incomplete answers
Clarity & Presentation	10 Marks	Clear, well-structured, and neatly presented answers	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Banker's Algorithm – Concepts

The Banker's Algorithm is a deadlock-*avoidance* method. Before granting resources, the system checks that the resulting state is **safe**, i.e. there is some order (a safe sequence) in which every process can obtain its maximum demand and finish. For n processes and m resource types it uses:

- **Available** – a vector of length m: the free instances of each resource type.
- **Max** – an n × m matrix: the maximum demand of each process.
- **Allocation** – an n × m matrix: the instances currently held by each process.
- **Need** – an n × m matrix: the remaining requirement of each process.

$$Need = Max - Allocation \qquad Total = \Sigma\, Allocation + Available$$

Safety algorithm:

1. Let Work = Available and Finish[i] = false for every process.
2. Find a process i with Finish[i] = false and Need[i] ≤ Work. If none exists, go to step 4.
3. Set Work = Work + Allocation[i] and Finish[i] = true; go back to step 2.
4. If Finish[i] = true for all i, the system is in a safe state; otherwise it is unsafe.

Question 1. Banker's Algorithm

Question: Considering a system with five processes P0 through P4 and three resources of type A, B, C. Resource type A has 10 instances, B has 5 instances and C has 7 instances. Suppose at time T0, the following snapshot of the system has been taken. (a) What will be the content of the Need matrix? (b) Is the system in a safe state? If yes, then what is the safe sequence?

Answer:

The Allocation and Max tables for this question were not included (they were given as an image that was not attached), so the Need matrix and safe sequence cannot be calculated. The method is:

1. Compute the Need matrix: Need = Max − Allocation (for example, Need[P0] = Max[P0] − Allocation[P0], element by element).
2. Compute Available = Total − Σ Allocation, using the totals given: A = 10, B = 5, C = 7.
3. Apply the safety algorithm starting with Work = Available, recording Work after each process finishes.
4. If all five processes finish, write the safe sequence in the order in which they were selected.

Question 2. Banker's Algorithm

Question: Consider the following snapshot for understanding Banker's Algorithm. Available = (2, 1, 0). Find: 1. Calculate the Need matrix. 2. Check whether the system is in a safe state. 3. Determine the total sum of each type of resource.

Process	Alloc A	Alloc B	Alloc C	Max A	Max B	Max C
P0	1	1	2	4	3	3
P1	2	1	2	3	2	2
P2	4	0	1	9	0	2
P3	0	2	0	7	5	3
P4	1	1	2	1	1	2

Answer:

1. Need Matrix

$$Need = Max - Allocation$$

- P0: Need = (4, 3, 3) – (1, 1, 2) = (3, 2, 1)
- P1: Need = (3, 2, 2) – (2, 1, 2) = (1, 1, 0)
- P2: Need = (9, 0, 2) – (4, 0, 1) = (5, 0, 1)
- P3: Need = (7, 5, 3) – (0, 2, 0) = (7, 3, 3)
- P4: Need = (1, 1, 2) – (1, 1, 2) = (0, 0, 0)

Process	Max (A B C)	Allocation (A B C)	Need (A B C)
P0	4 3 3	1 1 2	3 2 1
P1	3 2 2	2 1 2	1 1 0
P2	9 0 2	4 0 1	5 0 1
P3	7 5 3	0 2 0	7 3 3
P4	1 1 2	1 1 2	0 0 0

2. Safety Check

Work = Available = (2, 1, 0) and Finish = [F, F, F, F, F]. The processes are examined in order P0 → P4 in every pass; a process can run if Need ≤ Work, and when it finishes it returns its Allocation to Work.

Pass	Work at start	Process examined	Need	Need ≤ Work?	Action / Work after
1	(2, 1, 0)	P0	(3, 2, 1)	No	Skip (insufficient resources)
1	(2, 1, 0)	P1	(1, 1, 0)	Yes	P1 runs; Work = (2, 1, 0) + (2, 1, 2) = (4, 2, 2)
2	(4, 2, 2)	P0	(3, 2, 1)	Yes	P0 runs; Work = (4, 2, 2) + (1, 1, 2) = (5, 3, 4)
3	(5, 3, 4)	P2	(5, 0, 1)	Yes	P2 runs; Work = (5, 3, 4) + (4, 0, 1) = (9, 3, 5)
4	(9, 3, 5)	P3	(7, 3, 3)	Yes	P3 runs; Work = (9, 3, 5) + (0, 2, 0) = (9, 5, 5)
5	(9, 5, 5)	P4	(0, 0, 0)	Yes	P4 runs; Work = (9, 5, 5) + (1, 1, 2) = (10, 6, 7)

Finish after each step: P1 → [F, T, F, F, F]; P0 → [T, T, F, F, F]; P2 → [T, T, T, F, F]; P3 → [T, T, T, T, F]; P4 → [T, T, T, T, T].

All five processes can finish, so the system is in a **Safe State**.

Safe Sequence:

$$P1 \rightarrow P0 \rightarrow P2 \rightarrow P3 \rightarrow P4$$

P4 has Need = (0, 0, 0), so it can be run at any point after the first step; for example P1 → P4 → P0 → P2 → P3 is also a valid safe sequence. A safe state can have more than one safe sequence.
Check: the final Work = (10, 6, 7) equals the total number of instances of each resource, which confirms that all resources are returned when every process finishes.

3. Total Resources

$$Total = \Sigma\, Allocation + Available$$

- Resource A: Σ Allocation = 1 + 2 + 4 + 0 + 1 = 8; Total = 8 + 2 = 10
- Resource B: Σ Allocation = 1 + 1 + 0 + 2 + 1 = 5; Total = 5 + 1 = 6
- Resource C: Σ Allocation = 2 + 2 + 1 + 0 + 2 = 7; Total = 7 + 0 = 7

Resource	Allocated	Available	Total
A	8	2	10
B	5	1	6
C	7	0	7

Final Answer:

- Need matrix: P0 = (3, 2, 1), P1 = (1, 1, 0), P2 = (5, 0, 1), P3 = (7, 3, 3), P4 = (0, 0, 0)
- The system is in a safe state; safe sequence P1 → P0 → P2 → P3 → P4
- Total A = 10, Total B = 6, Total C = 7

Question 3. Banker's Algorithm

Question: We have a process table with an Allocation column (resources A, B, C), a Max column (maximum resources A, B, C), and an Available column (currently available resources). Find: (a) Construct the Need matrix. (b) Is the system in a safe state?

Answer:

The process table for this question was not included (image not attached), so the exact answer cannot be calculated.

$$Need = Max - Allocation$$

1. Subtract Allocation from Max, element by element, for every process to get the Need matrix.
2. Set Work = Available and Finish[i] = false for all i.
3. Pick any unfinished process with Need ≤ Work; set Work = Work + Allocation and mark it finished.
4. Repeat. If all processes finish, the state is safe and the order of selection is the safe sequence; if no unfinished process can proceed, the state is unsafe.

Question 4. Banker's Algorithm

Question: Using Banker's Algorithm, answer the following: i) How many resources of type A, B, C, D are there? ii) What are the contents of the Need matrix? iii) Find whether the system is in a safe state. If it is, find the safe sequence.

Answer:

The table containing Allocation, Max and Available was not included (image not attached), so the numerical solution cannot be calculated.

$$Need = Max - Allocation$$

$$Total = \Sigma\, Allocation + Available$$

1. (i) Add the Allocation column of every process for each of A, B, C, D, then add the Available vector to get the total instances.
2. (ii) Compute Need = Max − Allocation for each process and each of the four resource types.
3. (iii) Run the safety algorithm (Work = Available; pick Need ≤ Work; Work += Allocation) over all four resource types and list the sequence.

Question 5. Banker's Algorithm

Question: Answer the following using Banker's Algorithm: i) What is the content of the Need matrix? ii) Is the system in a safe state? iii) If a request from process P1 arrives for (0,4,2,0), can the request be granted immediately?

Answer:

The Allocation, Max and Available matrices were not included (image not attached), so the exact answer cannot be calculated.

Resource-Request Algorithm for P1 requesting Request[P1] = (0, 4, 2, 0):

1. If Request[P1] ≤ Need[P1], go to step 2; otherwise raise an error, because the process has exceeded its maximum claim.
2. If Request[P1] ≤ Available, go to step 3; otherwise P1 must wait because the resources are not available.
3. Pretend to allocate: Available = Available − Request; Allocation[P1] = Allocation[P1] + Request; Need[P1] = Need[P1] − Request.

4. Run the safety algorithm on the new state. If it is safe, grant the request immediately; if it is unsafe, restore the old state and make P1 wait.

Important Formulas – Exam Revision

Term	Formula
Need Matrix	Max – Allocation
Total Resources	Allocation + Available
Work	Available
New Work	Work + Allocation
Safe State	All processes can finish
Unsafe State	No complete safe sequence
Request check	Request ≤ Need, then Request ≤ Available